

DE Lab9
Hayden Fuller

Q1:

a)

$$V_{OH} = V_{CC}$$

$$V_{OL} = V_{CEsat} = 0.15V$$

$$V_{IL} = V_{BEon} = 0.65V$$

$$V_{IH}$$

$$I_C = (V_{CC} - V_C) / R_C = (V_{CC} - V_{out}) / R_C = (5 - 0.15) / 11k = 0.4409mA$$

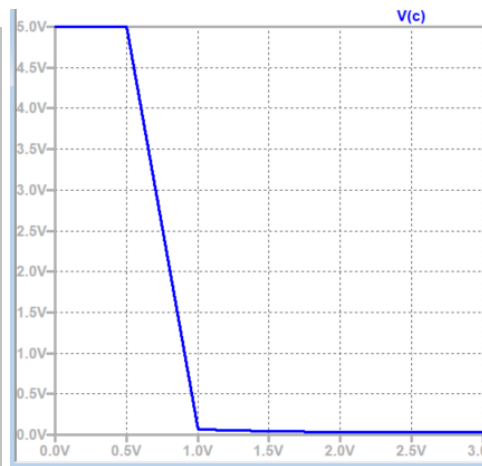
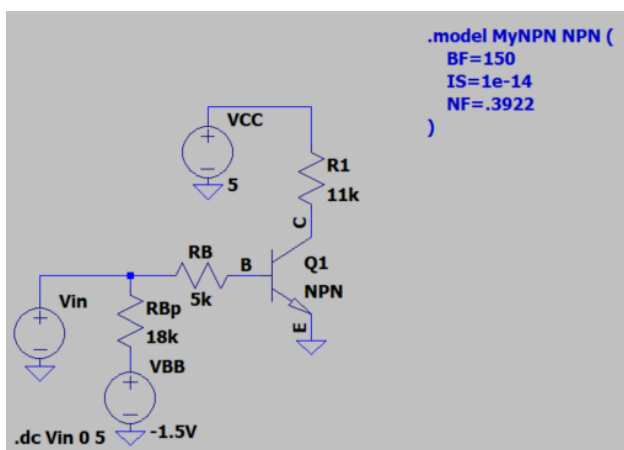
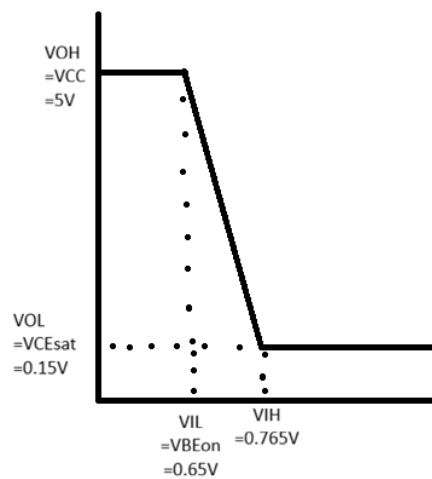
$$I_C = \beta I_B ; I_B = I_C / \beta = 0.4409mA / 150 = 2.939\mu A$$

$$I_B = (V_{in} - V_B) / R_B = 2.939\mu A = (V_{IH} - 0.75V) / 5k$$

$$V_{IH} - 0.75V = 2.939\mu A \cdot 5k = 14.69mV$$

$$V_{IH} = 0.75V + 0.01469V = 0.76469V$$

b)



It was difficult to figure out how to model the same transistor in LTspice, so this graph is relatively accurate in shape but shows different values at the points, $V_{IL} = 0.5V$, $V_{IH} = 1V$, and $V_{OL} = 0.065V$

c)

where $NM'H = 1.75V$:

I_{OH}

$$NM'H = V'_{OH} - V_{IH} = 1.75$$

$$V'_{OH} = NM'H + V_{IH} = 1.75V + 0.76469V = 2.514V$$

$$I_{OH} = (V_{CC} - V'_{OH}) / R_C = (5V - 2.514V) / 11k = 0.2259mA = 225.9\mu A$$

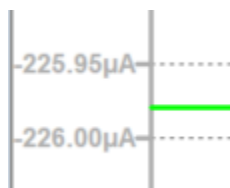
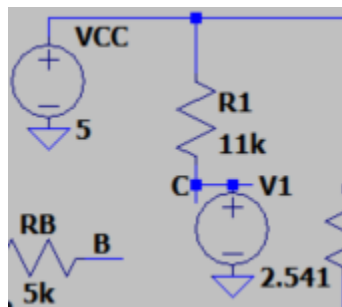
I_{IH}

$$I_{IH} = (V'_{OH} - V_{BB}) / R_{B'} + (V'_{OH} - V_B) / R_B = (2.514V - (-1.5V)) / 18k + (2.514V - 0.75) / 5k$$

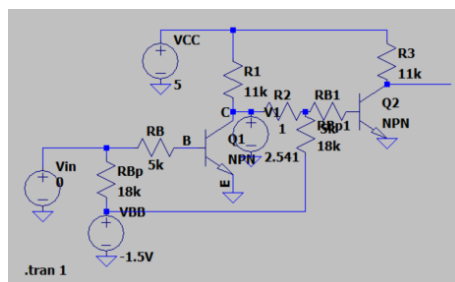
$$= 0.2230mA + 0.3529mA = 0.5759mA$$

$$N = V_{OH} / V_{IH} = 225.9\mu A / 575.9\mu A = 0.3922 < 1$$

cant power a second stage



$I'_{OH} = 226\mu A$



$I_{IH} = 577\mu A$

$N = 0.3916$

Because I was easily able to specify $NF = 0.3922$, I got very similar results in this test with nearly identical currents and N ratio.